

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

## PROFESSIONAL EXPERIENCE

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### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Performed mechanical assembly, electrical connections, and robot calibrations for new semiconductor production equipment from integration through inspection, resolving ~15% of initial issues independently.
- Followed detailed checklists and schematics to ensure correct procedures were followed during equipment installation, documenting adherence and any deviations.
- Tested equipment functionality with guidance from senior engineers and existing documentation, applying basic diagnostic techniques to identify and report on system performance.
- Responded to factory-reported issues by communicating with cross-functional engineering teams and actively participating in escalation conference calls to expedite problem resolution.
- Contributed to continuous improvement initiatives focused on reducing installation cycle time and associated costs through process analysis and feedback sessions.
- Provided accurate daily status reports, documented hours worked, and detailed non-conformances encountered during installation phases for project management oversight.
- Containerized core diagnostic and reporting utilities using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency across project sites.

## SKILLS

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**Installation & Assembly** – Mechanical Assembly, Electrical Connections, Robot Calibration, Troubleshooting, Tool Inspection, Schematics Reading

**Field Service & Support** – Customer Communication, Status Reporting, Documentation, Non-conformance Reporting, Escalation Calls, Problem Resolution

**Tools & Technologies** – MS Office Suite (Excel, PowerPoint), Jira, Confluence, Git, Linux Shell Scripting, AWS EC2, Docker

**Project Management** – Timeline Adherence, Budget Control, Quality Requirements, Process Improvement, Cycle Time Reduction

**Technical Background** – Mechanical Engineering, Electrical Engineering, Mechatronics, Equipment Installation, Semiconductor Manufacturing

## PROJECTS

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### Semiconductor Equipment Installation & Calibration Framework

- Developed a standardized framework for installing and calibrating new wafer fabrication equipment, incorporating **mechanical assembly** and **electrical connection** procedures.
- Integrated troubleshooting guides and diagnostic checklists directly into the installation process, enabling engineers to resolve approximately **20% of common issues** without escalation.
- Documented installation progress and system readiness using custom templates that fed into a central project tracking system, improving visibility for the installation task force.

### Automated Diagnostic Reporting for Production Tools

- Created Python scripts to automate the collection and initial analysis of diagnostic logs from installed equipment, reducing manual review time by ~40%.
- Built a **Flask**-based web interface to display key performance indicators and potential issues identified by the diagnostic scripts, accessible via internal company network.
- Containerized the reporting application using **Docker** for consistent deployment across different testing environments.

## EDUCATION

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### M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026



## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)